Elyas Chikhaoui

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Applied Mathematics & Machine Learning | Télécom Paris – Institut Polytechnique de Paris

End-of-study internship (6 months) – ML research / data science – available from March 2027

Work & Leadership Experience

INRIA Paris **Jan. 2026 – Jul. 2026**
Research Intern – Machine Learning & Symbolic Regression

- Designed and implemented **DragonSR**, a symbolic regression framework casting equation discovery as the optimisation of directed acyclic graphs (DAGs), built on top of DRAGON (Keisler et al., *JMLR* 2024): multi-channel representation with sub-expression sharing, feature-selection operators, least-squares channel aggregation (sparse, nested, polynomial-rational), bandit-driven search (Mutant-UCB).
- Noise-robust pipeline: Gaussian-process denoising, stochastic subsampling, variable augmentation with XGBoost feature selection, correlation-based objective and structural complexity penalty.
- Benchmarked on 27 equations (Nguyen, 9 physical laws, 9 Sentinel-2 remote-sensing indices with 10+ input variables) against **PySR** (state of the art), over 4 noise levels $\times$ 10 runs: DragonSR recovers formulas up to 10% noise where PySR recovers none (Hubble, ideal gas, Kepler, Leavitt, SAVI, EVI2…).
- **First author** of *DragonSR: Directed Acyclic Graph Search for Equation Discovery* (with J. Keisler and A. Charantonis), arXiv preprint forthcoming; code and results released (GitHub, Zenodo).

Independent Commodity Intelligence Services (ICIS) – LexisNexis Risk Solutions **Jul. 2025 – Dec. 2025**
Quantitative Analyst Intern – Gas Markets

- Extended a GAMS-based LP model of the European gas network: proposed and implemented a triangle-function approach to model LNG sendout at terminals, replacing a continuous real-valued discharge with a time-distributed carrier unloading profile.
- Developed 15+ analytics charts on the proprietary ICIS portal (SQL) covering location spreads, forward curves and pipeline utilisation; built a data pipeline ingesting missing actuals from the ENTSOG API to populate flex-supply stress indicators; built from scratch an interactive map of daily and forecast European gas flows and prices (Plotly, Dash).
- Refactored and modularised the outage processing pipeline under a new data model, reducing per-module runtime from $\sim$3 h to $\sim$20 min (Python, SQL).

Sopra Steria **Jul. 2024 – Aug. 2024**
Summer Intern – Digital Unit, Aeroline Division

- Contributed to *Bhubble*, an Angular-based web app for team and project management (GitLab integration, polls, tech tracking).
- Built a Python automation script for ticket migration (Tuleap $\rightarrow$ Spira); worked in an Agile environment with TypeScript, REST APIs and PostgreSQL.

Télécom IA & Unaite – Student AI Associations **Jun. 2024 – Jul. 2025**
President & Co-founder

- President of *Télécom IA*: organised AI events and technical workshops with InstaDeep and Kyutai.
- Co-founded *Unaite*, a national alliance of AI student associations from 6 top French schools, 100+ members (Télécom Paris, École Polytechnique, ENPC, CentraleSupélec, HEC, ESSEC).

Education

Télécom Paris | Institut Polytechnique de Paris **Sep. 2023 – Jun. 2027**
MSc in Engineering – Applied Mathematics

- **Coursework:** Stochastic Calculus, High-Dimensional Probability, Machine Learning, Computer Science, Sociology.
- **Exchange semester at AGH University of Kraków** (Poland, Fall 2026): Stochastic Algorithms, Digital Image Processing & Vision Systems, GPU programming (CUDA / OpenCL), Modelling & Optimisation with GAMS, Soft Computing, Semiconductors in Neuromorphic Computing.

Lycée Henri IV **Sep. 2020 – Jul. 2023**
Preparatory Classes (MPSI & MP)

- Intensive preparation in mathematics and physics for the nationwide competitive entrance exams to French *Grandes Écoles*.

Skills & Interests

Languages French (native), English (fluent, Cambridge Linguaskill: C1), Polish (conversational)
Technical Python (PyTorch, scikit-learn, Pandas, NumPy, Plotly, Dash), SQL / PostgreSQL, GAMS, Git, LaTeX
Interests Karate (black belt), Football